AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter Monitoring & Health Effects

Issue 2026-09-06 | Entry window 6 September 2026 | 15 records screened in scope

15 RECORDS IN SCOPE (21 EXCLUDED)	**10** EFFECT ESTIMATES EXTRACTED	**10** WITH A HUMAN HEALTH ENDPOINT	**4** ON THE SENSING / EXPOSURE AXIS	**0** TRIAL UPDATES (SUNDAY ZERO)

Signal of the day

The $PM_{2.5}$ concentration–response slope is steepening as concentrations fall — and it has already clawed back a fifth of China's clean-air mortality benefit.

A multi-county time-series across the Sichuan Basin (*Int Arch Occup Environ Health*, 2014–2022, extended two-stage design with longitudinal meta-regression) finds the excess risk of cardiorespiratory mortality per 10 μg/m^3 $PM_{2.5}$ moved **non-monotonically** across the phases of the Clean Air Action: **0.7% (95% CI 0.6–0.9) in 2014–2016, 0.4% (0.3–0.6) in 2017–2019, then 1.4% (1.2–1.6) in 2020–2022** ($P < 0.001$ for the temporal pattern). Decomposed against a counterfactual, falling concentrations avoided **52,831** cardiorespiratory deaths while the drift in the exposure–response function *added* **10,041** — so roughly 19% of the achieved benefit was given back, concentrated in those aged 65 and over. The mechanism is unresolved and the paper does not claim one; changing composition at lower mass, a shifting susceptible fraction, and residual confounding by the 2020–2022 period itself are all live. The instrumentation consequence is sharp and independent of which explanation wins: as the mean falls, the health-relevant contrast shrinks while the per-unit risk grows, so a network's useful resolution has to improve just to hold epidemiological power constant. A monitor calibrated to be adequate at 60 μg/m^3 is not adequate at 25, and mass-only calibration cannot detect a composition-driven slope change at all.

What changed today

- **PubMed deposits on Sundays — the 5 September zero was not a weekend rule.** This window is a Sunday and PubMed returned **9 health-axis and 1 sensing-axis** hits to the harvester, and 10 and 13 to the connector. Saturday's both-ways-verified zero was therefore a property of that day, not of weekends. **ClinicalTrials.gov is the opposite case** and its weekend zero *is* structural: the no-term control returns **0 studies updated registry-wide** on 6 September, as on 5 September and on 29/30 August.
- **Fifteen records and ten effect estimates — the forest plot returns after a one-day gap.** 5 September carried no estimate with a confidence interval at all. Today eight of the ten come from just two papers, so the panel shows within-study precision far more than between-study agreement, and is captioned as such.
- **Solid fuel outranks ambient PM inside the mixture.** In CHARLS ($n = 10{,}273$ for cardiometabolic multimorbidity), quantile g-computation puts every co-exposure up one quantile together for an **OR of 1.650 (1.375–1.979)**, and identifies **indoor solid-fuel use as the primary contributor** — ahead of PM_1, $PM_{2.5}$, PM_{10} and heatwave. Household combustion keeps outperforming outdoor mass in mixture weights whenever both are measured together.
- **Day-to-day *change* in $PM_{2.5}$, not level, predicts infarction.** The Zhongshan case-crossover (1,795 admissions) reports **+7.89% (0.86–15.40) odds per 10 μg/m^3 of lag0–1 increment**, rising to 17.90% for ST-elevation infarction. A first-difference exposure metric is cheap for a sensor network to compute and is insensitive to the constant offset that dominates low-cost PM error — the rare case where an uncalibrated node still carries the signal.
- **A low-cost sensor is the exposure instrument in a health study, and the result is a warning.** The Douala survey (1,721 individuals, 344 households) drives an instrumental Logit from an OC 300 Dust Particle Laser and reports effects of **0.231 (PM_{10}), 0.122 (PM_5) and 0.018 ($PM_{2.5}$)** — monotonically *decreasing* with decreasing size, which inverts the usual deposition argument. An uncorrected optical instrument apportioning mass across bins it cannot resolve is the parsimonious explanation, and it is what a co-location arm would have settled.
- **A clean null: exercise benefit is not modified by industrial emissions.** CLHLS, 16,954 adults aged 65+, six years: habitual exercise **HR 0.835 (0.773–0.902)**, county industrial SO_2 **HR 1.001 (0.874–1.146)**, and **no significant interaction** on any of four pollution indicators. County-level soot and SO_2 tonnage is a weak exposure surrogate, so this bounds the design more than the hypothesis.
- **The connector's sensing axis is now mostly noise.** 13 hits, **10 rejected outright** and only two new to this issue. Free-text `"remote sensing"` and `"exposure assessment"` pulled in phycotoxin hydrography and a football-heading index; particle terms must be ANDed in.

Corpus analytics

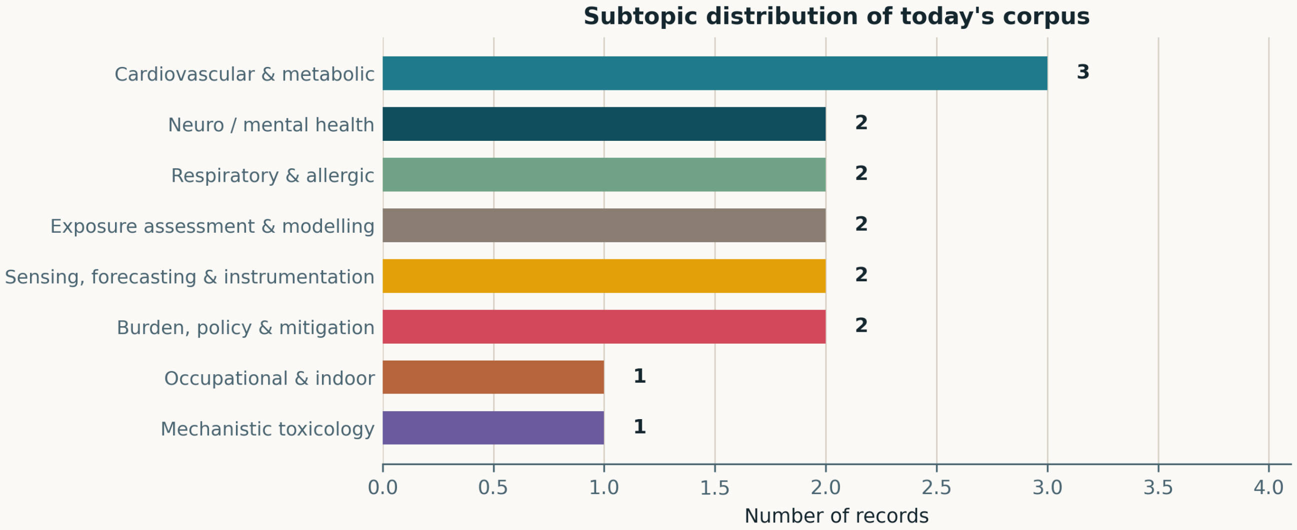
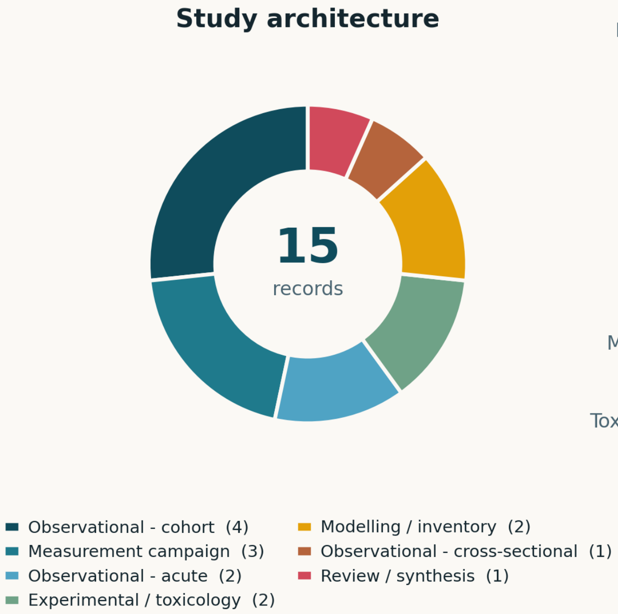


Fig. Subtopic assignment of the 15 in-scope records. Eight of the ten clusters are occupied; *Reproductive & developmental* and *Other clinical endpoints* are empty. *Cardiovascular & metabolic* leads with three, and no cluster exceeds three — an unusually flat distribution for an issue of this size, and the reason no single theme carries the day.



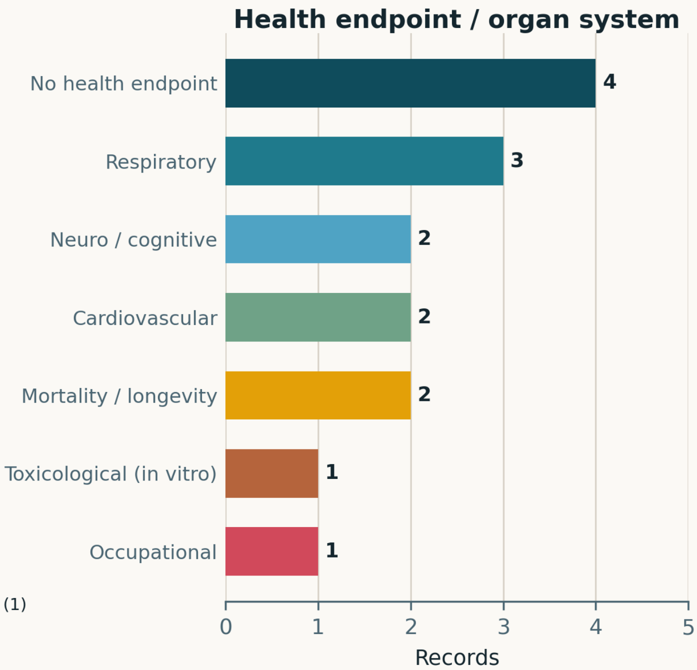


Fig. Left — study architecture. Observational cohorts account for 4 of 15 and measurement campaigns for 3; the two *Modelling / inventory* entries are the SAQIN inference network and the California wine-grape regression. Right — health endpoint. **Ten records carry a human health endpoint**, the inverse of 5 September, when only two did. Respiratory leads at three; the five without a health endpoint are the two sensing records, the exposome review, the crop-yield preprint and the in-vitro toxicology.

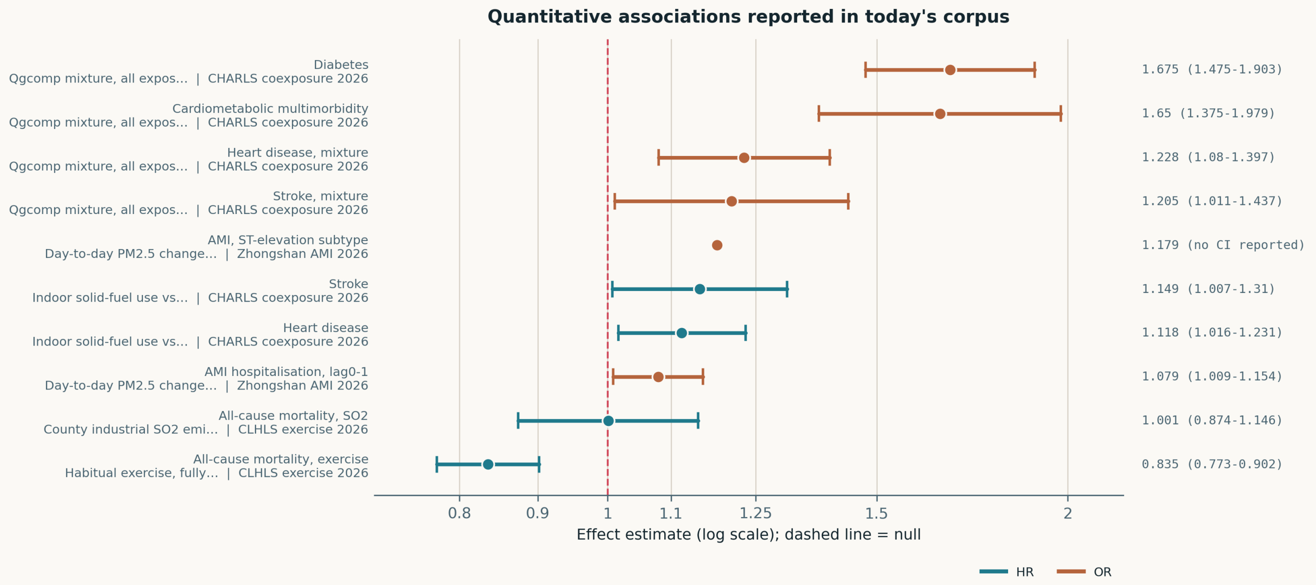


Fig. All ten quantitative estimates in the corpus, on a common logarithmic axis. **The exposure contrasts are not mutually comparable** — four are quantile g-computation mixture ORs in which every co-exposure moves together, two are per-10 $\mu g/m^3$ day-to-day $PM_{2.5}$ increments, two are binary solid-fuel contrasts, and two are non-particulate (habitual exercise, county SO_2 tonnage). Read direction and precision only. Eight of the ten come from two papers, so the visual clustering is within-study, not corroboration. The ST-elevation subtype estimate is plotted as a point because the source reports no interval for it.

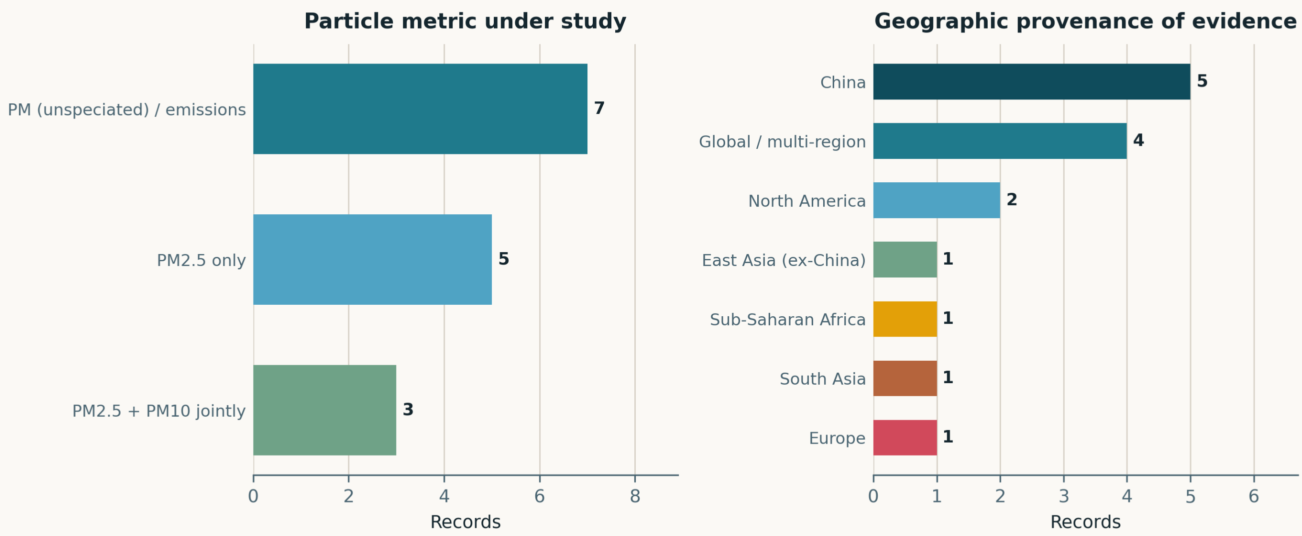


Fig. Left — particle metric. *PM (unspeciated) / emissions* is the largest bin at 7, which overstates the field's precision: it holds county SO_2 and soot tonnage, occupational coal dust by job history, dosed reference materials and self-reported cooking fuel. Only 8 of 15 records measure a size-resolved particle metric at all. Right — geographic provenance. China supplies 5. The *Global / multi-region* bar is a fallback bucket, not a finding: one genuine multi-region review plus two records whose setting is unstated and one bench study.

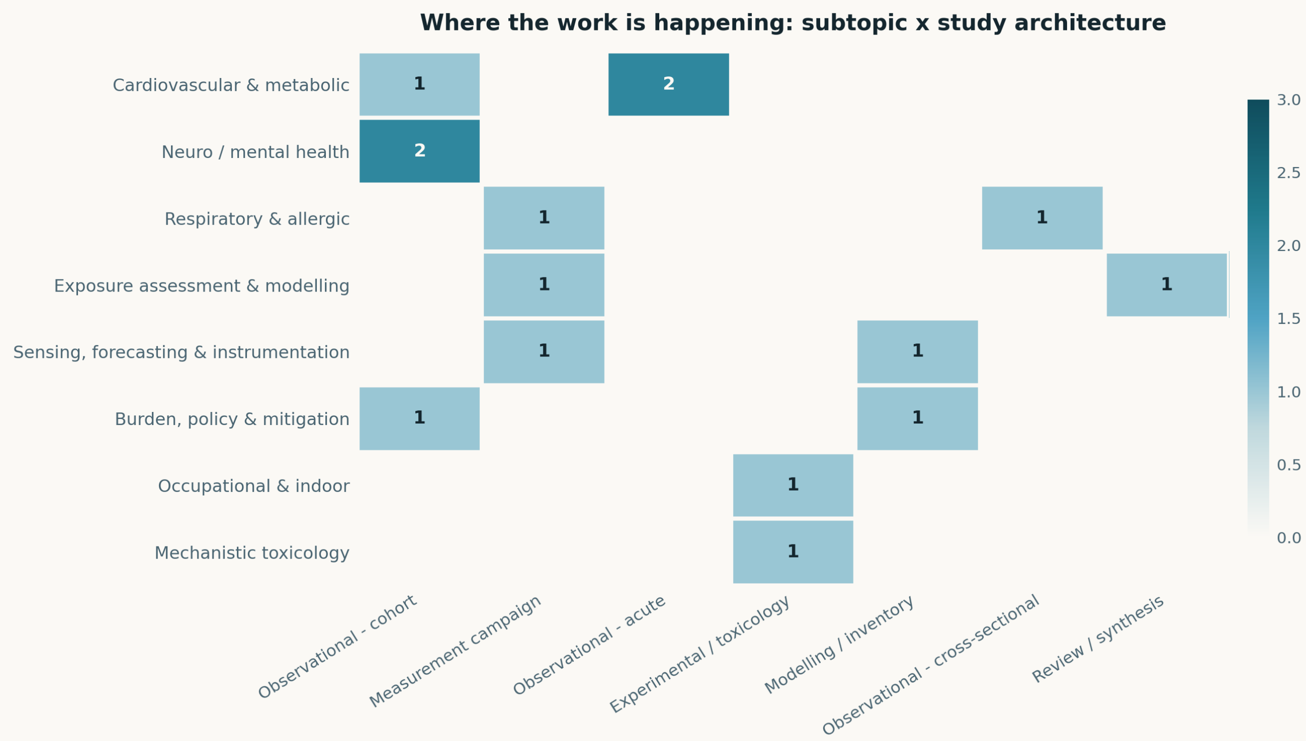


Fig. Subtopic × study-architecture cross-tabulation. Twelve of 56 cells are occupied and none holds more than two records, so this is a maximally dispersed day. The structural point is that every two-record subtopic *except* neuro spreads its records across two different architectures. Note also that *Exposure assessment* is not siloed today: the Douala record sits in that row and carries a respiratory endpoint, which is the linkage this cross-tabulation usually shows missing.

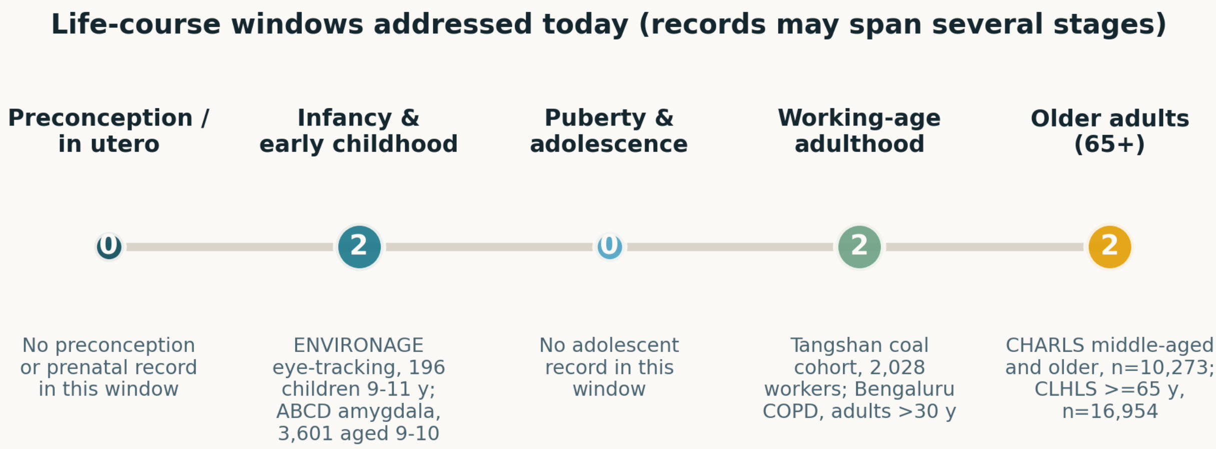


Fig. Life-course windows addressed; counts are non-exclusive. Three windows carry two records each; the in-utero and adolescent windows are empty, and the reproductive and developmental cluster is absent from this issue entirely.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Cardiovascular & metabolic — 3 records

Sichuan Basin CAA 2026 Int Arch Occup Environ Health

Temporal changes in short-term $PM_{2.5}$–cardiorespiratory mortality associations under the Clean Air Action

Multi-county time-series, Sichuan Basin 2014–2022, extended two-stage design with county-specific estimates pooled and longitudinal meta-regression on annual excess risk. ER per $10\,\mu g/m^3$ traced **0.7% (0.6–0.9)** → **0.4% (0.3–0.6)** → **1.4% (1.2–1.6)** across the three policy periods, $P < 0.001$ for the pattern; scenario decomposition gives 52,831 deaths averted by concentration decline against 10,041 added by the risk change, and the late rise held in both sexes and was larger at ages 65+. The threat to the inference is that the final period is 2020–2022: pandemic-era changes in care-seeking, mortality coding and time–activity are not separable from a true slope change in this design. Still the most consequential result here — a falling mean does not imply a falling per-unit risk, and networks must resolve smaller contrasts as policy succeeds.

doi: 10.1007/s00420-026-02237-z

Zhongshan AMI 2026 Front Public Health

Daily changes in $PM_{2.5}$ and acute myocardial infarction hospitalisation

Individual-level time-stratified case-crossover, 1,795 AMI admissions to one tertiary hospital in Zhongshan 2020–2024, 5,877 control days matched on weekday within calendar month; residential $PM_{2.5}$ from the CHAP 1 km product; conditional logistic regression adjusted for temperature, humidity and holidays. A 10 $\mu g/m^3$ lag0–1 *increment* gave **+7.89%** (**0.86–15.40**) odds, with 22.58% in women, 22.16% at ages >65, 10.85% in the cold season and 17.90% for STEMI; age modified the effect ($P_{int} < 0.05$). Robust to co-pollutants except NO_2, which is the honest weakness — traffic co-exposure is not excluded. Single-hospital catchment limits generalisability, and the 1 km modelled field is assigned to address, not measured. The transferable idea for sensing is the first-difference metric: it cancels the additive bias that dominates low-cost PM error.

doi: 10.3389/fpubh.2026.1911027

CHARLS coexposure 2026 Int Arch Occup Environ Health

Indoor solid fuels, outdoor PM and heatwave vs. cardiometabolic multimorbidity

Five CHARLS waves 2011–2020, four disease cohorts (CMM $n = 10{,}273$; diabetes 9,893; heart disease 9,536; stroke 10,466), time-varying Cox plus a cumulative risk index and quantile g-computation. Single-exposure: solid fuel **HR 1.118** (**1.016–1.231**) for heart disease and **1.149** (**1.007–1.310**) for stroke; PM_1, $PM_{2.5}$ and PM_{10} were the most consistent risk factors ($P < 0.001$). Mixture ORs: **CMM 1.650** (**1.375–1.979**), **diabetes 1.675** (**1.475–1.903**), with solid fuel carrying the largest weight. Fuel use is self-reported and correlates with income, education and rurality, so the weight assigned to it is the estimate most exposed to residual confounding. Outdoor PM is modelled, not measured, at residence.

doi: 10.1007/s00420-026-02232-4

Neuro / mental health 2 records

ENVIRONAGE eye-track 2026 EBioMedicine

Early-life air pollution and oculomotor markers of cognitive workload

196 children aged 9–11 from the ENVIRONAGE birth cohort completed eye-tracking paradigms 2021–2023; residential BC, NO_2 and $PM_{2.5}$ from a high-resolution spatiotemporal model combining fixed-site monitors, land use and dispersion; multiple linear regression then BKMR distributed-lag models across windows. After multiple-testing correction the surviving associations are *recent* exposure with saccadic outcomes — per IQR of last-month BC **–10.97°/s** (**–18.67, –3.26**) and NO_2 **–10.81** (**–18.13, –3.49**) saccade velocity — plus first-trimester and first-1000-days windows on the same outcome. Pupil diameter, fixation count and attention-bias outcomes did not survive. $n = 196$ against many windows and pollutants means the surviving hits are the ones to trust and the null ones are underpowered rather than negative. Notable for sensing: the effect attaches to last-month exposure, a window a personal or neighbourhood monitor can actually measure.

doi: 10.1016/j.ebiom.2026.106469

ABCD amygdala 2026 (preprint) bioRxiv (preprint)

Air pollution, early-life adversity and amygdala volume in preadolescents

3,601 ABCD Study children aged 9–10, annual residential $PM_{2.5}$ total mass plus 15 components, linear mixed-effects adjusted for sociodemographics, co-pollutants and imaging covariates. Adversity predicted general, internalising and externalising symptoms; **$PM_{2.5}$ moderated the adversity–externalising association**, with the highest symptoms in children high on both, and smaller basolateral paralaminar volumes tracked symptoms within the high-adversity group. Cross-sectional, so the mediation reading the framing invites is not supported; annual average exposure at residence cannot speak to the timing the developmental argument needs; and with 16 exposure terms the component results want a correction the abstract does not describe. Not peer reviewed. The component panel is the part worth watching — it is the kind of speciated contrast that mass-only networks cannot support.

doi: 10.64898/2026.08.31.748438

Respiratory & allergic 2 records

COPD anaemia metals 2026 Int J Chron Obstruct Pulmon Dis

Air-pollution-associated intracellular metals, immune gene expression and erythrocyte indices

61 COPD patients and 10 controls; annual PM_{10}, $PM_{2.5}$, NO_2 and NO_x by land use regression; metals by ICP-MS and single-cell ICP-MS in PBMCs and red cells; PBMC RNA-seq. $PM_{2.5}$ was associated with higher RBC count ($\beta = 0.0314 \times 10^6/\mu$L, 0.0066–0.0561), red cell distribution width, and with intracellular chromium, zinc and cadmium; those metals in turn tracked *ALPL*, *CXCR1* and *PI3* expression, and anaemic patients showed a distinct signature. Ten controls against 61 cases and a cross-sectional design make every link correlational, and LUR annual means cannot support a claim about cellular metal kinetics. Read it as an assay-development result: single-cell ICP-MS as an internal-dose readout is the transferable piece, and it is the kind of endpoint a speciating monitor could eventually be validated against.

doi: 10.2147/COPD.S626348

Bengaluru COPD 2026 Indian J Community Med

Prevalence and risk factors of COPD in an urban poor locality

Community-based cross-sectional survey, adults over 30 in an urban poor locality of Bengaluru, PPS sampling to a calculated 1,560, spirometry-confirmed diagnosis. Prevalence **10.38%** overall, 14.4% in men and 7.1% in women, with 74.06% at moderate-to-severe stage; multivariate logistic regression retained age, smoked and smokeless tobacco, fuel type, biomass exposure, environmental tobacco smoke and indoor air pollution. No air measurement of any kind — every exposure is a questionnaire item, so the “indoor air pollution” association is a self-report contrast and the effect sizes are not reported in the abstract. Value here is the spirometry-confirmed prevalence in a population that ambient monitoring networks systematically under-serve.

doi: 10.4103/ijcm.ijcm_145_25

Exposure assessment & modelling 2 records

Douala households 2026 Spat Spatiotemporal Epidemiol

Particulate concentration and household respiratory health in urban Cameroon

On-site campaign in Douala with an OC 300 Dust Particle Laser — a low-cost portable optical sensor — paired with a survey of 1,721 individuals in 344 households sited near the measurement points; instrumental Logit to address endogeneity of exposure. Reported effects on respiratory disease were **0.231 for PM_{10}, 0.122 for PM_5 and 0.018 for $PM_{2.5}$**, and unsealed roads and traffic density dominated the PM levels. The size ordering is the problem: effect falls monotonically as particles get smaller, the reverse of what deposition physics predicts, and the most economical explanation is an uncorrected nephelometric instrument distributing mass across bins it cannot separate — no co-location against a reference monitor is described, and no humidity correction. Included because it is a rare health study built on a low-cost node, and because it shows precisely what the missing co-location arm costs.

doi: 10.1016/j.sste.2026.100836

Paediatric exposome 2026 Pediatr Res

Assessing environmental exposures in children: toward a comprehensive exposome approach

Narrative review of exposure-assessment tooling for paediatrics — questionnaires, biomarkers, wearable and fixed sensors, geospatial indicators and pooled databases — alongside longitudinal cohort design, multi-omics integration and methods for mixtures and critical windows. No systematic search protocol, no pooled estimate, and the closing argument is professional advocacy rather than evidence. Useful only as a citation anchor for the claim that single-pollutant assessment is the limiting factor in child environmental epidemiology, and for its framing of sensors as one instrument among several rather than the exposure itself.

doi: 10.1038/s41390-026-05460-z

Sensing, forecasting & instrumentation — 2 records

SAQIN 2026 Neural Netw

Long-short-term decoupled residual learning for high-resolution spatio-temporal air quality inference

Inference — not forecasting — of high-resolution concentration fields from sparse reference stations. The model splits a static branch (semantic segmentation, remote-sensing context, population density, elevation) from a dynamic branch (multi-head self-attention over geographic relationships and inter-pollutant coupling), then treats inference as residual correction anchored to multiple stations, fused by distance-weighted averaging across stations. That last step is the substantive contribution: it removes the Voronoi-partition discontinuities that single-anchor schemes produce, yielding a continuous field. Reported as superior on large-scale real-world data with zero-shot cross-domain generalisation, but the abstract names no dataset, no station count, no RMSE and no baseline, so none of it is checkable here. The architecture is directly applicable to dense low-cost networks, where the anchor stations are the few reference-grade nodes.

doi: 10.1016/j.neunet.2026.109575

Wildfire CHON PM2.5 2026 Environ Res

FT-ICR MS with rule-based prediction and QSAR for nitrogen-containing organics in wildfire $PM_{2.5}$

Molecular-level characterisation of $PM_{2.5}$ from a temperate coniferous forest wildfire in South Korea, wildfire versus post-wildfire periods. GC-MS gave nitrated polycyclic aromatic compounds **3.4× higher during the fire** (**0.122 vs 0.036 ng/m^3**), with 2-nitrofluorene then 3-nitrofluoranthene dominant; FT-ICR MS showed a relative shift toward CHON formulae, and rule-based SOA prediction from 12 VOC precursors matched limonene-derived products most often. QSAR LC_{50} values of 1.0, 1.6 and 8.3 mg/L for β-pinene-, α-pinene- and isoprene-derived compounds are ranked hazard, not measured toxicity, and the authors say so — the assignments are formula-level, so isomers with very different toxicity are pooled. One fire, one forest type; the compositional shift is the result, the hazard ranking is a hypothesis.

doi: 10.1016/j.envres.2026.125576

Occupational & indoor exposure — 1 record

Tangshan CWP IGF1 2026 Front Public Health

Dust exposure, IGF1 and autophagy in coal workers' pneumoconiosis

Retrospective analysis of 2,028 employees at one Tangshan coal enterprise (2023–2024) with a paired SiO_2-stimulated macrophage model. Logistic regression retained early initial working age, dust-exposure duration, smoking, white cell count and AST/ALT ratio; patients and stimulated macrophages both showed raised IGF1, altered LC3/P62 and increased PI3K/AKT/mTOR phosphorylation. Two weaknesses matter. Exposure is job history, not a measured respirable dust concentration, so no exposure–response can be estimated; and the cell model uses crystalline silica while the cohort's hazard is coal mine dust — these are different particles with different mechanisms, and the paper's own language stays correlative. The white-cell and liver-enzyme "risk factors" are as plausibly consequences of disease as causes.

doi: 10.3389/fpubh.2026.1899729

Mechanistic toxicology — 1 record

Retinal CRM 2026 Biochem Biophys Rep

Certified particulate reference materials in primary mouse retinal cells

Two certified environmental particulate reference materials applied to primary mouse retinal cells, plus repeated ocular-surface exposure in mice. Effects on DCFDA fluorescence and ATP-dependent viability were condition-dependent and concentrated in glial conditions; neither α-tocopherol nor N-acetylcysteine gave a significant rescue, and glial coculture did not improve retinal ganglion cell survival. In postnatal mice topical particulate was associated with inner retinal thinning; in adult Thy1-CFP mice the within-animal comparison was null and only a *post hoc* between-animal pooling reached significance (**–14.4 cells, –28.1 to –0.7,** $P = 0.040$). The authors state plainly that this establishes neither particle transport to the retina nor a causal oxidative mechanism nor any ambient risk — an unusually disciplined set of negative caveats, and the reason this is tier C rather than a finding. Ocular-surface dosing is not inhalation, and certified reference material is not ambient PM.

doi: 10.1016/j.bbrep.2026.102758

Burden, policy & mitigation — 2 records

CLHLS exercise 2026 Front Public Health

Exercise, county industrial air pollution and all-cause mortality in older adults

16,954 CLHLS participants aged 65+, 23 provinces, three waves over six years, Cox models with time-varying covariates and sequential adjustment. Habitual exercise **HR 0.835 (0.773–0.902)**; county industrial SO_2 **HR 1.001 (0.874–1.146)** after adjustment; soot indicators null; and **no significant interaction** between exercise and any of the four pollution indicators. The null is about the exposure metric as much as the hypothesis: annual county soot and SO_2 emission and removal tonnage is an administrative aggregate, not a concentration, and it carries neither $PM_{2.5}$ nor any within-county gradient. Exercise is self-reported and reverse causation from subclinical illness is not addressed. Reported here as a bounded null, not as evidence against interaction.

doi: 10.3389/fpubh.2026.1894835

CA wine-grape smoke 2026 (preprint) arXiv (preprint)

Quantifying impacts of wildfire smoke on crop yields in California

Statistical regression of wine-grape yield on particulate matter alongside climate predictors at three Northern California sites. PM entered as a significant predictor jointly with temperature at two of the three, with final models reaching $R^2 = 0.79$ **at Sonoma and 0.66 at Napa** — read by the authors as a shift from climate-controlled to wildfire-and-climate-controlled yields. Three sites, yield as an annual aggregate, and no separation of smoke PM from background or from the heat and radiation anomalies that accompany fire weather; with that many candidate predictors, an R^2 of this size is weak evidence of a specific PM pathway. Included as the only record this window treating PM as an economic rather than a health exposure, and because the smoke-attribution problem it dodges is the same one source-resolved monitoring exists to solve. Not peer reviewed.

doi: 10.48550/arXiv.2609.04422

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
ABCD amygdala 2026	Air Pollution, Early Adversity, and Amygdala: Environmental Correlates of Psychopathology in Preadolescents	*bioRxiv (preprint)*	Prospective cohort	Annual residential $PM_{2.5}$ total mass plus 15 components	United States
ENVIRONAGE eye-track	Residential early-life air pollution exposure and oculomotor markers of cognitive workload in children: a prospective birth cohort study	*EBioMedicine*	Prospective cohort	Modelled residential BC, NO2 and $PM_{2.5}$; windowed, per IQR	Europe
Cardiovascular & metabolic					
CHARLS coexposure	Association of coexposure to indoor solid fuels, outdoor air pollution, and heatwave with the risk of Cardiometabolic Multimorbidity among middle-aged and older Chinese adults	*Int Arch Occup Environ Health*	Prospective cohort	PM1, $PM_{2.5}$ and PM_{10} with solid-fuel use and heatwave, Qgcomp mixture	China
Sichuan Basin CAA	Temporal changes in short-term $PM_{2.5}$-cardiorespiratory mortality associations and implications for the health benefits of the clean air action in the Sichuan Basin, 2014-2022	*Int Arch Occup Environ Health*	Time-series	Ambient $PM_{2.5}$, county daily means; ER per 10 ug m-3	China
Zhongshan AMI	Daily changes in fine particulate matter are associated with acute myocardial infarction hospitalizations: an individual-level time-stratified case-crossover study in Zhongshan, China	*Front Public Health*	Case-crossover	Day-to-day $PM_{2.5}$ change, CHAP 1 km, residence-assigned	China
Respiratory & allergic					
Bengaluru COPD	Prevalence and Risk Factors of Chronic Obstructive Pulmonary Disease (COPD) in an Urban Poor Locality of Bengaluru, South India - A Cross-sectional Study	*Indian J Community Med*	Cross-sectional (multistage)	Self-reported cooking fuel, biomass and indoor smoke; no measurement	India
COPD anaemia metals	Air Pollution-Associated Intracellular Metals, Immune Gene Expression, and Erythrocyte Indices in COPD with Anemia	*Int J Chron Obstruct Pulmon Dis*	Repeated-measures biomarker	Annual PM_{10}, $PM_{2.5}$, NO2, NOx by land-use regression	Not stated
Mechanistic toxicology					
Retinal CRM	Environmental particulate reference materials increase DCFDA fluorescence and reduce ATP-dependent viability in primary mouse retinal cells	*Biochem Biophys Rep*	Experimental / toxicology	Two certified environmental particulate reference materials, dosed	Laboratory
Occupational & indoor					
Tangshan CWP IGF1	Effects of dust exposure on IGF1 and autophagy related to coal workers' pneumoconiosis	*Front Public Health*	Experimental / toxicology	Cumulative occupational coal dust by job history; SiO2 in vitro	China
Sensing, forecasting & instrumentation					

First author	Focus	Journal	Design	Metric	Region
SAQIN	Long-short-term decoupled residual learning for high-resolution spatio-temporal air quality inference	*Neural Netw*	Machine-learning model	Criteria-pollutant fields inferred from sparse reference stations	Not stated
Wildfire CHON $PM_{2.5}$	Integrating FT-ICR MS with Rule-Based Prediction and QSAR to Resolve Nitrogen-Containing Organic Compounds and Their Toxicity in Wildfire-Derived $PM_{2.5}$	*Environ Res*	Measurement campaign	Wildfire $PM_{2.5}$ by GC-MS and FT-ICR MS; nitrated PAC in ng m-3	South Korea
Exposure assessment & modelling					
Douala households	What is the impact of the concentration of particulate matter on the respiratory health of households in urban Cameroon?	*Spat Spatiotemporal Epidemiol*	Observational exposure study	PM_{10}, PM5, $PM_{2.5}$ by OC 300 Dust Particle Laser, low-cost portable	Sub-Saharan Africa
Paediatric exposome	Assessing environmental exposures in children: toward a comprehensive exposome approach	*Pediatr Res*	Narrative review	Survey of questionnaires, biomarkers, sensors and geospatial indicators	Global
Burden, policy & mitigation					
CA wine-grape smoke 2026	Quantifying Impacts of Wildfire Smoke on Crop Yields in California	*arXiv* (*preprint*)	Modelling / inventory	Regulatory PM as a yield predictor alongside temperature covariates	United States
CLHLS exercise	Exercise is associated with lower all-cause mortality among older adults: a national cohort survival analysis with county-level industrial air pollution indicators	*Front Public Health*	Prospective cohort	County annual industrial soot and SO2 emissions and removals	China

Provenance and method

Entry window **6 September 2026**, a single day, run against `[Date - Entry]`. **PubMed** (NCBI E-utilities) returned **9 health-axis** and **1 sensing-axis** hits to the harvester; the **PubMed connector**, run independently on both axes, returned **10** and **13**, contributing three in-scope records the harvester missed. The sensing-axis surplus is almost entirely false positive — free-text `"remote sensing"` and `"exposure assessment"` matched marine phycotoxins, sports biomechanics, volcanic petrology and MODIS evapotranspiration — and **10 of those 13 were rejected**, a third in-scope hit being one the health axis had already returned. **Europe PMC** (`CREATION_DATE`) returned 6 raw, **Crossref-by-ISSN** 11 raw, **arXiv** 1 inside the local recency screen. **OpenAlex** returned HTTP 429, unchanged since 28 July. **Consensus** returned three hits, all out of window on Crossref `created` date (2019, 2022, 2023) — a nil for the eighth consecutive run, and the absence of a date post-filter remains the cause. **Scholar Gateway** was queried but its response exceeded the reply budget and was not consumed; the abstracts already carried the quantities needed, so no record depended on it. **ClinicalTrials.gov** posted **no updates registry-wide** on 6 September — verified with a no-term control, as on 5 September and 29/30 August — so the trial zero is a posting-calendar artefact of the weekend and is not evidence about PM trial activity. Twenty-eight raw records deduplicated to 23 unique on PMID → DOI → normalised title; none had been shipped in a prior issue. Twenty-one records were screened out with a logged reason, chiefly non-atmospheric *Environ Sci Technol* chemistry, built-environment and energy papers with no particle metric, ozone- and VOC-only studies, one corrigendum, and the connector's sensing-axis false positives. Subtopic, design, particle-metric and geography labels are assigned from title, abstract and keywords; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. *Global / multi-region* in the geography panel is a fallback bucket that absorbs unstated and non-geographic settings. Effect estimates are transcribed verbatim and use heterogeneous contrasts — see the forest-plot caption. All DOI links resolve to the publisher of record; the arXiv record is cited by its DataCite DOI. Abstracts remain the copyright of their respective publishers.